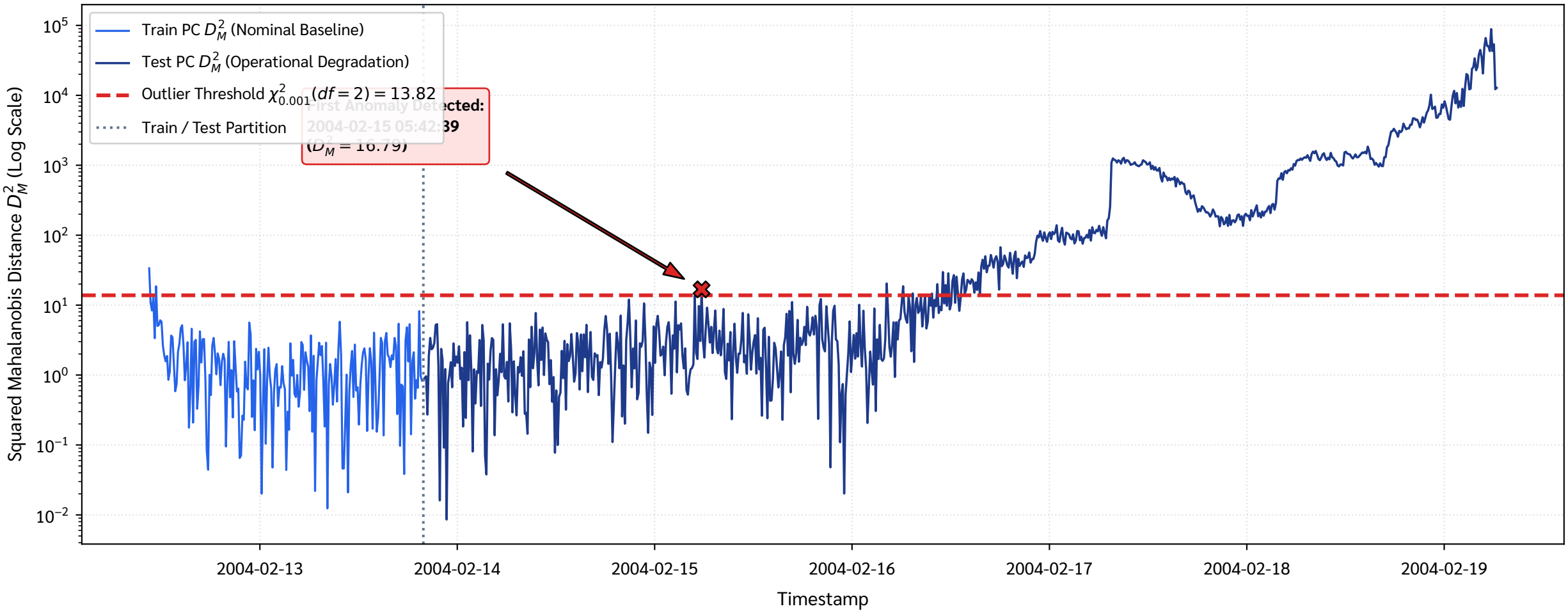
Figure 6: Latent-Variable Mahalanobis Distance (T^2) Anomaly Detection Timeline
Train PC D_M^2 (Nominal Baseline)
Test PC D_M^2 (Operational Degradation)
Outlier Threshold $\chi^2_{0.001}(df = 2) = 13.82$
Train / Test Partition
Squared Mahalanobis Distance D_M^2 (Log Scale)
10^5
10^4
10^3
10^2
10^1
10^0
10^{-1}
10^{-2}
2004-02-13
2004-02-14
2004-02-15
2004-02-16
2004-02-17
2004-02-18
2004-02-19
Timestamp